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To cite this article: Katarzyna Samborska-Goik & Leonard I. Wassenaar (2025) Reimagining the Kendall plot: using $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ of nitrate and advanced machine learning to improve N pollutant source classification, *Isotopes in Environmental and Health Studies*, 61:3, 310-335, DOI: [10.1080/10256016.2025.2467863](https://doi.org/10.1080/10256016.2025.2467863)

To link to this article: <https://doi.org/10.1080/10256016.2025.2467863>



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STABLE ISOTOPE TRACER & METHODOLOGY



Reimagining the Kendall plot: using $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ of nitrate and advanced machine learning to improve N pollutant source classification

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ABSTRACT

Nitrate (NO_3^-) pollution is a serious water quality issue in many countries due to contamination of lakes, rivers, and aquifers by intensive agriculture practices and inadequate wastewater management. Nitrate pollution and associated cultural eutrophication are anticipated to increase worldwide, highlighting the need to control and reduce nitrogen pollution. The stable isotope ratios of nitrate ($\delta^{15}\text{N}$, $\delta^{18}\text{O}$) are widely used as tracers of nitrogen pollution sources. The primary technique for identifying nitrate sources has been the longstanding Kendall boxplot, despite improved methods using Bayes' theorem and the R language for estimating source fractions using hydrogeochemical context, N source data and expert assessment. This article improves the classification of aqueous nitrate sources using comprehensive published stable isotope data for nitrate from four known pollutant types and applying machine learning algorithms. AI modelling results reveal improved source depictions and offer a robust statistical framework for identifying N pollution sources. This is essential given the increased published data sources and the need for better-informed water quality management strategies.

ARTICLE HISTORY

Received 13 August 2024
Accepted 16 January 2025

KEYWORDS

Nitrate isotopes; water pollution; ^{15}N ; ^{18}O ; machine learning; neural network; decision tree; classification

1. Introduction

Nitrogen pollution of terrestrial and marine waters from agricultural, industrial and wastewater sources is a serious global issue [1–4]. Nitrogen fertilizer use increased from $< 10 \text{ Tg N ha}^{-1}$ in 1950–77 Tg N ha^{-1} by 2016, with a forecast of 111.5 thousand tonnes by 2022 [5,6]. Excess nitrogen applications from various sources have compromised the water quality of many aquatic ecosystems, resulting in environmental health issues, including eutrophication, algal blooms, fish mortality, and increased N_2O production and pose a health risk for human infants [7,8]. Reducing nitrogen inputs into aquatic environments is crucial for safeguarding environmental sustainability and human health [9–11].

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 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/10256016.2025.2467863>.

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The tracing of aqueous nitrate sources is possible by analyzing the stable isotopes of nitrogen ($\delta^{15}\text{N-NO}_3$) and oxygen ($\delta^{18}\text{O-NO}_3$), which may exhibit distinct isotopic characteristics by N source, although isotopic overlap may occur [12–16]. For example, the $\delta^{18}\text{O}$ of nitrate from precipitation tends to be greater than +30 ‰ [14, 17–22]. In contrast, the $\delta^{15}\text{N}$ of organic fertilizers (e.g. manure, wastewater) are typically more positive than precipitation and inorganic fertilizers, ranging between +5 and +25 ‰ [23–25]. Synthetic N fertilizers have low $\delta^{15}\text{N}$ values close to 0 ‰ as they are derived from atmospheric N_2 via the Haber–Bosch process [23,26] but have high $\delta^{18}\text{O}$ values [27].

1.1. Nitrate isotope classification

The $\delta^{18}\text{O}$ alone cannot efficiently differentiate between nitrate origins from atmospheric sources and nitrification or nitrate fertilizers due to the wide range of $\delta^{18}\text{O-NO}_3$ values in atmospheric deposition overlapping with those from nitrification or nitrate fertilizers. Therefore, the most widely used approach to identifying nitrate sources in water involves using a dual $\delta^{15}\text{N-NO}_3$ vs $\delta^{18}\text{O-NO}_3$ biplot, comparing the isotopic ranges to potential nitrogen sources (Figure 1). This methodology, commonly referred to as a Kendall Plot [14], has been widely adopted in terrestrial and aquatic systems [28–38].

Using a Kendall isotopic biplot (or boxplot) has become ubiquitous in classifying N sources. However, there is a risk of over-simplification; for example, Venkiteswaran et al. [39] observed that surface waters rarely show N derived from precipitation because its nitrate concentrations are so low and both biotic and abiotic processes quickly alter the isotopic composition. Therefore, simplified box biplots do not consider

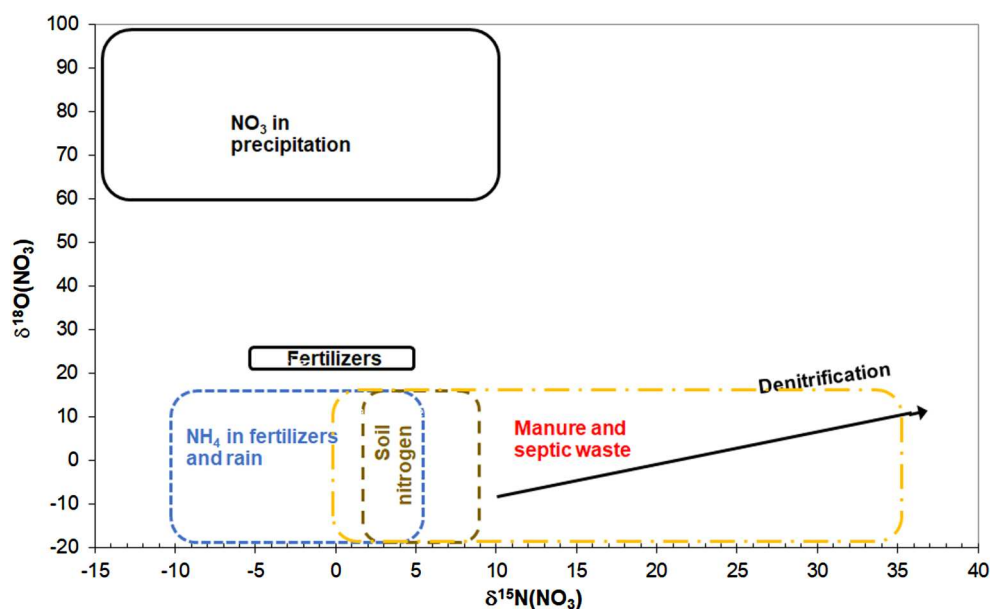


Figure 1. The original ‘Kendall Box Plot’ [14] illustrated ranges of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ data of nitrate sources. The arrow indicates a typical 2:1 slope for denitrification although slopes as low as 1:1 are not uncommon [189].

the various processes that can substantially modify the isotopic composition of aqueous nitrate. Accordingly, improved N source depictions are needed.

Nevertheless, the dual stable isotope approach to nitrate source identification provides a significant advantage by increasing the odds of accurately identifying sources, especially in scenarios where N and O isotopic ranges overlap, such as for fertilizers and soil nitrogen. Balancing quantitative information with qualitative assessments is crucial, drawing upon an increasing body of source data and expertise for N source identification. When N and O isotopic overlaps occur, mixing processes from different sources combined with intricate hydrogeochemical analyzes will be essential. Additional isotopic tracers, such as boron, have been suggested [40–42]. Additionally, $\Delta^{17}\text{O}$ serves as a tracer for identifying nitrates. In recent decades, the triple-oxygen isotopic approach, $\delta^{18}\text{O}$ and $\Delta^{17}\text{O}$ [43], revealed that atmospheric nitrate has high enrichment in ^{17}O from ozone interactions. The $\Delta^{17}\text{O}$ values of atmospheric nitrate are positive within the range of 17–35 ‰ [44–47], whereas the $\Delta^{17}\text{O}$ values of nitrates originating from other terrestrial sources is 0 ‰ [26,48].

1.2. Mathematical mixing models

Mathematical models are employed for source identification and estimating the percentage contribution of sources based on observed isotopic compositions. Linear mass balance mixing models like IsoSource [49] and Bayesian methods such as SIAR, SIMMR, or MixSIAR [50–52] are commonly used and assume conservative mixing. Linear mixing models are solved analytically in a dual-isotope tracer approach when dealing with a few sources [53,54]. However, in scenarios with many sources, numerical solutions like IsoSource or Bayesian models are usually more appropriate [55].

IsoSource is particularly effective when the number of N sources is limited to $n + 1$ of the isotopic tracers. This model has been widely applied in determining soil N sources, water sources in plant species [56] and nitrate sources in rivers and groundwater [57,58]. Bayesian models have also been used to estimate the contribution of nitrate sources in surface water [59–62]. IsoSource and Bayesian models generally produce similar results [63]. Still, the Bayesian model does not require an exact match between the number of sources and tracers and can incorporate prior information to validate the modelling outcomes.

1.3. Global datasets

The successful application of stable isotope methods for identifying nitrate sources is facilitated by the continuous publication of isotope datasets containing observations from various known sources of dissolved compounds and water bodies. Expanded datasets are critical in advancing our understanding of nitrate sources by providing comprehensive insights from diverse environmental sources. The Global Network of Isotopes in Rivers (GNIR) at the International Atomic Energy Agency (IAEA) is particularly significant [64]. Data contributions from researchers, institutions, and monitoring stations continuously update GNIR isotope databases. The data from GNIR is conveniently accessed from the Water Isotope System for Data Analysis, Visualization, and Electronic Retrieval (WISER).

1.4. AI methods

The foundations of artificial intelligence (AI) were established in the 1930s by visionaries like Alan Turing [65] and Norbert Wiener [66]. Machine learning is adept at addressing traditional statistical challenges like regression, classification, decision-making, and clustering and is distinguished by the problems it tackles and the methodologies it employs. AI involves the analysis of large amounts of intricate and unstructured data [67] and using robust algorithms and computational resources [68,69]. In the environmental sciences, machine learning algorithms function independently of explicit mathematical equations and may not rely on pre-existing hydrological or environmental background information. Instead, AI algorithms excel at unveiling intricate relationships between sets of predictors and predicted values. This data-centric approach facilitates the extraction of valuable insights and generates precise predictions [70]. However, it is essential to acknowledge that high predictive accuracy can result in overfitting, where models capture noise or irrelevant patterns in training datasets, potentially compromising the ability to generalize to unseen data or the complexity of the hypothesis [71,72]. The outcomes of machine learning models are challenging to interpret without a comprehensive understanding of the underlying environmental domain, potentially resulting in incorrect conclusions [73–75].

AI methods such as machine learning and deep learning models have increasing application in environmental studies, particularly the evaluation of water quality and nitrate contamination [76–78]. In the area of stable isotope-related investigations, machine learning algorithms are still rather limited [79], and have focused on the forecasting of $\delta^{18}\text{O}$ and $\delta^2\text{H}$ values in precipitation, groundwater and the oceans [80–83]. Yang et al. [84] were instrumental in introducing machine learning algorithms to forecast the stable isotopic composition of nitrate in water. In their research, 16 hydrochemical variables were employed to predict the nitrate and stable isotopic composition of surface waters of the Wen-Rui Tang River. Additionally, Xie et al. [85] conducted a study to establish connections between hydrogeochemical predictors and the stable isotope composition of nitrate in the Yangtze River. Their approach was a fusion of Bayesian mixing models and machine learning algorithms. In summary, the application of machine learning algorithms towards the isotopic composition of nitrate is a rapidly growing area of research.

This study is the first to use machine learning algorithms as classifiers on extensive nitrate-N stable isotope known source datasets. With increasing stable isotope datasets from diverse nitrate origins like precipitation, soil nitrogen, waste and fertilizers, a wealth of data is suitable for effectively training machine learning models or generating new datasets. This presents an opportunity to enhance the accuracy and reliability of source prediction models. Moreover, it is noteworthy that numerous studies of nitrate source identification continue to rely on basic Kendall biplot diagrams, which poses the question: Is this basic biplot still appropriate, and does it outperform classifications generated by AI algorithms? This requires a discussion about the advantages of algorithmically derived outcomes and the importance of gathering source N data and disseminating results for potential training of models in the future. With well-established models, samples can be classified at high precision and retrained over time as new data is gathered, enabling a continuous refinement and enhancement of the classification and visual depiction process.

2. Materials and methods

This study used $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ data for nitrate obtained from the literature and online databases that were determined to be derived from four types of isotopically distinctive sources: (1) atmospheric deposition (precipitation and dry deposition), (2) inorganic fertilizers, (3) organic fertilizers, municipal wastewater, and (4) natural unpolluted soil nitrogen. The query was conducted using online tools and bibliographic databases, including searches of university repositories such as those of Brown University and AGH University, as well as resources like Google Scholar, ResearchGate, and specific databases, including the Nitrate Isotope Database for Meteoric Waters, Surface Waters, Soil Waters, and Groundwaters [86]. The queries relied on keywords such as nitrates, isotopes, meteoric water, fertilisers, identification, and sources. Data compilations were limited to sources examining both isotopes, $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$, and identifying them as validated nitrate sources. Data were extracted from tables, text, and images. Since the machine learning algorithms used for classification required isotope data, only isotope values, data sources, and labels were retained for further analysis.

While emphasis has been placed on evaluating the performance of models using metrics like accuracy, sensitivity, precision, recall, mean squared error (MSE), and root mean squared error (RMSE) [87–89], the quality of the underlying isotopic data is of utmost importance and too often disregarded. Hence, constructing AI models must thoroughly vet the underlying dataset (e.g. garbage in, garbage out). Thus, data cleaning is an essential step in modelling [90–92], which involves rectifying issues like data incompleteness, sample outliers, duplication, inconsistency, typographical mistakes, analytical errors, and mislabelling. After expert data cleaning, users should be capable of asking inquiries like: (1) Do the isotope data display coherence? (2) Do the isotope data adhere to the designated principles of the domain? (3) Do the isotope data substantiate or challenge the characteristics of the model? (4) Are there recognizable and useful isotopic patterns within the data? [93]

Another practice that can improve data quality is transformations, which may include log-normalization, standardization, and discretization, which particularly influence methods that rely on Euclidean distance-based algorithms such as k -nearest neighbors or k -means, clustering or regression models that are prone to labelling bias [94,95]. Moreover, in scenarios where high-quality source isotope data are scarce, improving the reliability of AI models built on such datasets requires cross-validation and data augmentation. The latter uses tactics to artificially expand the dataset and enhance the existing information without changing its labels. Another technique is bootstrapping, which involves creating ‘new’ samples by randomly picking data from the original dataset with replacement, irrespective of the data’s underlying statistics [96]. Despite the limitations, like sensitivity to outliers and assumptions concerning sample independence [97], bootstrapping is essential for the development and validation of machine learning models and serves various purposes, including data augmentation for training, estimating confidence intervals for model predictions, and handling imbalanced class data [98–101]. Determining the optimal number of ‘new’ samples to generate has significance in bootstrapping methodology. Early recommendations suggested creating approximately 100–500 new samples [102,103]. But in the era of advanced computing power, the generation of thousands of new samples is readily done. As a general guideline, the number of new

samples should be adequate to approximate a quasi-normal distribution of the estimator [104], often falling within the range of 5,000–10,000. Bootstrapping is a straightforward, efficient technique for bolstering machine learning algorithms, providing valuable insights into parameter estimation, model validation, and uncertainty quantification.

2.1. Data sources

This study utilized published $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ data on the isotopic composition of nitrates from known sources. The dataset for precipitation/snow/deposition nitrate was derived from Sebestyen et al. [86], with a sample size of $n = 352$. Additional precipitation isotope data from other sources were as follows: [105] ($n = 94$), [106] ($n = 88$), [107] ($n = 52$), [108] ($n = 48$), [109] ($n = 48$), [110] ($n = 34$), [111] ($n = 31$), [112] ($n = 29$), [113] ($n = 28$), [114] ($n = 17$), [115] ($n = 10$), [30] ($n = 8$), [116] ($n = 7$), [117] ($n = 5$), [118] ($n = 3$), [119] ($n = 3$), and [120] ($n = 2$), resulting in $n = 859$ samples for precipitation.

The dataset for soil-derived nitrate was sourced from: [86] ($n = 236$), [121] ($n = 54$), [122] ($n = 26$), [123] ($n = 13$), resulting in $n = 329$ data points for soil nitrate.

The dataset for nitrate derived from manure, organic fertilizers, and wastewater was based on [124] ($n = 54$), [125] ($n = 28$), [117] ($n = 7$), [126] ($n = 4$), [120] ($n = 3$), [119] ($n = 3$), and [118] ($n = 1$) for $n = 100$ for animal and wastewater nitrate.

Finally, nitrate from inorganic fertilizers was characterized by data from multiple sources: [124] ($n = 19$), [127] ($n = 14$), [128] ($n = 13$), [129] ($n = 6$), [117] ($n = 4$), [30] ($n = 2$), [130] ($n = 1$), and [131] ($n = 1$), resulting in a total of $n = 60$.

After scrutinizing these varied datasets for missing values, mislabelling, and duplicates, anomalies and given the disparity in the number of samples across different nitrate sources and to improve the model's performance, the data underwent resampling to a total of 1000 samples specifically for the training dataset. Subsequently, the data was split into test and training datasets in a ratio of 30:70. All datasets used are summarized in the Supplementary Materials. It is important to emphasise that isotopes were measured using different analytical methods, each with varying degrees of analytical error. To obtain the largest possible dataset, neither method was prioritized. Nonetheless, future modelers may exclude certain samples and establish new classification boundaries as deemed appropriate.

2.2. Supervised classification methods

Supervised classification is commonly used in machine learning and data mining [132]. It requires a model capable of predicting class labels based on data features. Various classification algorithms, like support vector machines, decision trees, Bayesian techniques, and neural networks, leverage the available data to perform these tasks with a certain level of confidence [133]. Classification metrics like accuracy and area under the curve are assessed on the test dataset, which is reserved during training, to gauge how effectively the model categorizes the input feature vector. To delineate clear isotopic boundaries between nitrate sources and to classify the samples, robust machine learning algorithms are employed. Initially, logistic regression and linear support vector machines (SVM) were tested to assess the linear isotopic boundaries between atmospheric nitrogen, organic fertilizers, wastewater, and soil nitrogen. However, due to the significant overlap of

inorganic fertilizers with other N sources, additional non-linear solutions were explored. These included Naïve Bayes classifier, radial basis function (RBF) support vector machines (SVM), *k*-nearest neighbors (kNN), multi-layer perceptron (MLP) neural networks, and decision trees. Only the three most useful of these algorithms are described in detail below: (i) the Naïve Bayes classifier, (ii) the multi-perceptron neural network, and (iii) the decision tree classifier. These three algorithms demonstrated the strongest performance in handling imbalanced and overlapping isotopic data. Furthermore, the depiction of the boundaries between the nitrate sources by all models was easier to interpret visually and improved the classic Kendall biplot approach.

2.3. Naïve Bayes algorithm

The Naïve Bayes algorithm is the most straightforward probabilistic classifier that calculates probabilities by tabulating the frequencies and combinations of values within a given dataset. It leverages Bayes' theorem and assumes all variables are independent given the class variable. However, this assumption can be a drawback as it does not match real-world scenarios [134]. Consequently, irrelevant and noise-contaminated features have the same importance as significant features, potentially affecting the classification accuracy. Despite this, the Naïve Bayes algorithm shows swift learning capabilities across various well-controlled classification tasks including clinical medicine [135]. Further, the Naïve Bayes classifier is capable of handling uncertain data [136–138] as well as small datasets [139,140].

2.4. Multilayer perceptron

Frank Rosenblatt developed the first artificial neural network (ANN), known as the perceptron in 1959 [141]. This model comprised an input layer with weighted inputs, a summation function, and an activation function that determines the neuron's output. The perceptron was specifically designed for binary classification tasks, operating under the assumption that the data was linearly separable. Overall, the architecture of ANN is inspired by biological neurons [142] and consists of nodes and weighted connections that influence outcomes [143]. The adjustment of these weights is referred to as the training process and involves three key layers: the input layer (representing input signals), the hidden layer (akin to brain function), and the output layer (representing the signal output). Additionally, three matrices – input weight, neuron weight, and output weight matrices – govern the connections within the network. The input weight matrix manages the connections from the input layer to the hidden layer, the neuron weight matrix controls connections between adjacent layers, and the output weight matrix oversees connections from the hidden layer to the output layer of the network. The effective design and training of a multilayer perceptron (MLP) necessitate critical considerations, including determining the number of hidden layers, selecting the number of neurons in each hidden layer, identifying suitable activation functions, and optimizing the learning rate. Furthermore, parameters such as batch size, regularization techniques, optimizer choices, initialization methods, dropout rates, and training epochs significantly impact the efficacy of MLP design and training [144–147].

2.5. Decision tree classifier

The output of the decision tree classifier is a flowchart-like tree structure or a partitioned area where each region corresponds to a specific class. This classifier is particularly effective at pattern recognition, especially in non-linear class separability instances. The training set facilitates the learning process, which consists of input patterns (features) and their corresponding labels (classes). The algorithm evaluates which features (or attributes) in the data are most informative for classifying the instances. Subsequently, a root node is established based on the most informative feature. In contrast, internal nodes represent tests on attributes, branches indicate the possible outcomes of these tests, and leaf nodes signify class labels. The classification rules are determined by the path from the root node to the leaf [148,149]. By utilizing the partitioned regions or decision tree, automatic classification of previously 'unseen' samples can be achieved [150]. This method is widely used due to its ease of implementation, rapid processing, and relatively straightforward conceptual framework compared to other machine learning algorithms. Decision tree classifiers can attain accuracy comparable to and, in some cases, superior to that of other classification methods, all while requiring minimal computing memory [151]. Also noteworthy is the handling of different input data, such as nominal, numeric and textual, as well as datasets with missing data or outliers [152,153].

3. Results and discussion

Figure 2 depicts the synthesized literature $\delta^{15}\text{N}$ vs $\delta^{18}\text{O}$ data of nitrates derived from four known sources. For comparison, we superimposed the commonly used Kendall 'boxes' which aimed to identify key N sources. A visual inspection reveals that global nitrate isotope source signals deviate widely from the classic Kendall box plot fields. This revealed

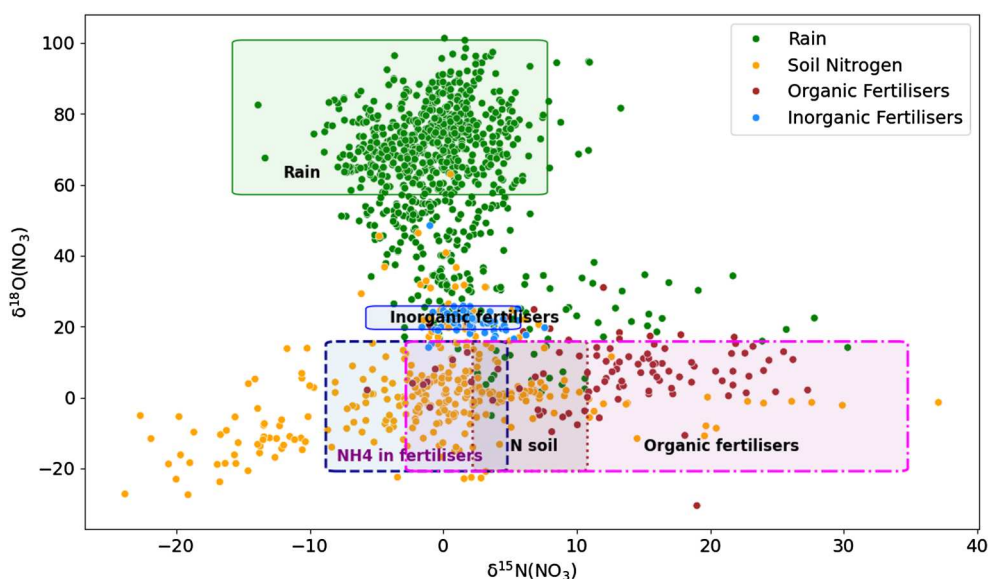


Figure 2. Global dataset ($n = 1348$) biplot of four nitrate $\delta^{15}\text{N}$ vs $\delta^{18}\text{O}$ sources from the literature (see Supplemental Materials) plotted against the classic Kendall box fields representing nitrate sources.

that previous interpretations made using this simple box-plot approach are likely subject to high uncertainty, which could be reduced by applying sophisticated AI methods and a better understanding of the hydrological and chemical context. This raises the prospect of an AI-generated biplot that could be more reliable. At the outset, several assumptions had to be made regarding the published nitrate isotope source data, specifically that they accurately represented these four sources and were not a result of mixing or alteration by abiotic or biotic processes like denitrification. Various machine learning techniques enabled the delineation of boundaries between the four nitrate sources: rain and deposition, soil nitrogen, organic fertilizers, and inorganic fertilizers. These trained AI models can serve as ongoing tools for identifying unknown samples when they become available.

Below, we describe the results of two non-linear methods in greater detail, namely Naïve Bayes and decision tree classifiers. These methods are efficient and straightforward and yield good accuracy with visually appealing classification fields that are easily grasped.

Incorporating isotope data on inorganic fertilizers required an evaluation of non-linear models, particularly as fertilizer $\delta^{15}\text{N}$ data often aligns with that of precipitation (Figure 2). The analysis of data and testing machine learning algorithms such as logistic regression and linear SVM revealed that establishing coherent linear boundaries between these two sources was impossible due to significant isotopic overlap. Subsequently, the non-linear Naïve Bayes classifier was tested. Concerning this algorithm, a crucial parameter that can influence the results is prior information. Within this model, one option involved setting up priors, like the ratio between the number of classes and the number of samples or by adjusting this parameter to reflect no prior information. Figure 3 shows the Naïve Bayes classifier results considering no prior information, which gave better performance. The plot is informative and accurately portrays N source classification with an accuracy of 89.6 % and a precision score of 89.3 %. Moreover, the plot was aesthetically pleasing as the classification fields were visible, exhibiting improved smooth boundaries compared to the Kendall boxes in Figure 1. Together with plots illustrating the boundaries between sources, a confusion matrix is also presented. A confusion matrix is a vital tool in machine learning employed to evaluate a classification model's performance. It summarizes the model's predictive capabilities by displaying the number of true positives, true negatives, false positives, and false negatives. This enables a nuanced understanding of how well the model performs across different classes.

Figure 4 illustrates the results of the multilayer perceptron (MLP) classifier. This method required several critical parameters: hidden layer sizes, activation method, solver, parameter alpha, learning rate, and maximum iterations. The MLP is a feedforward neural network [154,155] wherein data flows unidirectionally from the input to the output layer. The basic architecture of an MLP comprises only three layers: the input layer, the hidden layer, and the output layer. The hidden layer receives weighted inputs from the input layer and transmits data from the preceding layer to the subsequent one. In general, the depth of a network – defined as the total number of layers minus one (excluding the input layer) – indicates the number of non-linear transformations applied between the successive layers. Meanwhile, the dimensionality of a hidden layer, represented by the number of hidden neurons, is referred to as its width.

For the MLP architecture with one hidden layer, a generalized solution for all Parity-N cases requires neurons $n = N + 1$, where N is the parity number as well as the number of neurons in the hidden layer [156]. The classification of a Feedforward Neural Network

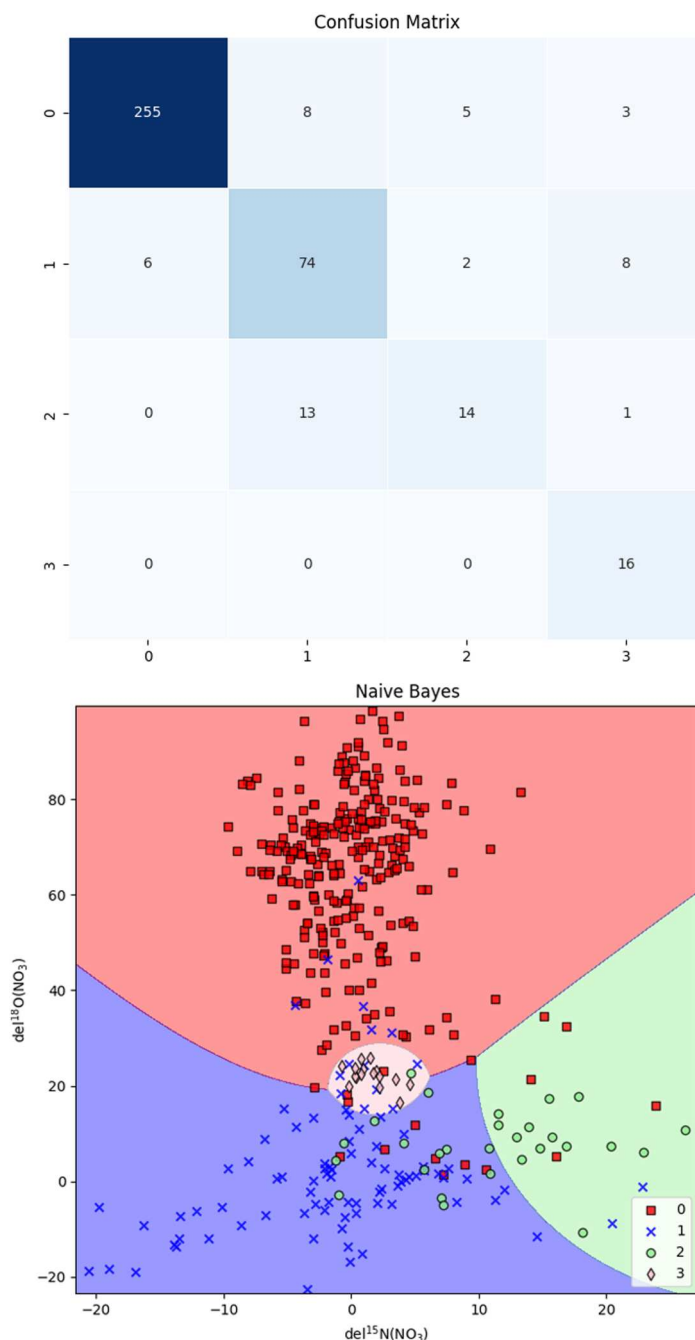


Figure 3. Classification of nitrate N sources using the Naive Bayes model. The red field represented precipitation sources, and the red squares indicated the test data for this source. The purple field represented soil nitrogen, with purple crosses representing the test data. The green field represented organic fertilizers with wastewater, and green circles indicate the test data for this source. The pink diamonds represent data for inorganic fertilizers, and the pink field pertains to the isotopic extent for two isotopes for this source. On the left is the confusion matrix for the model. The labels 0, 1, 2, and 3 corresponded precipitation, soil nitrogen, organic fertilizers with wastewater, and inorganic fertilizers, respectively.

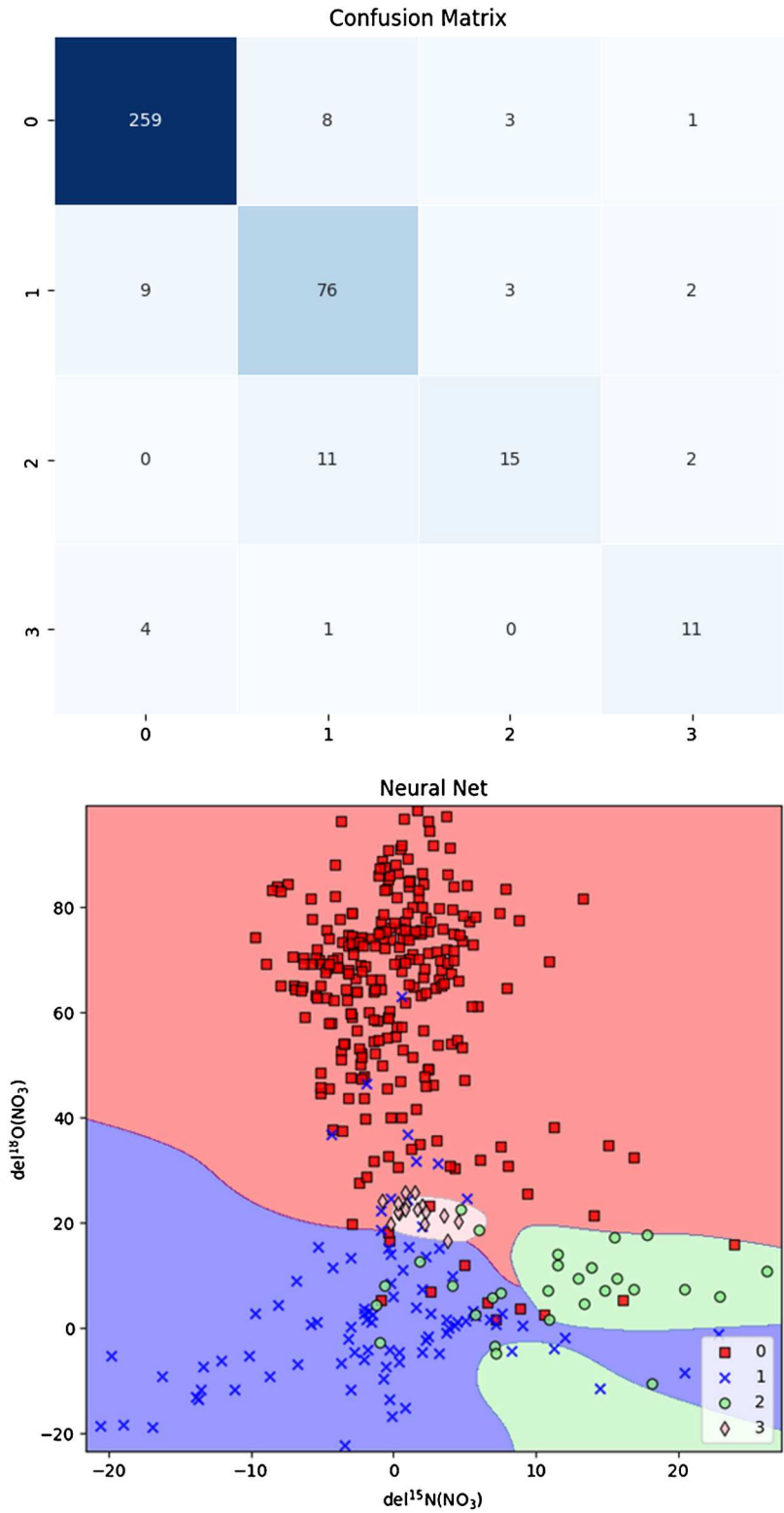


Figure 4. The results of the MLP classification model of nitrate isotope N sources use the same legend for colours and markers as in the previous figure.

(FFNN) architecture as 'deep' is subject to debate; however, architectures with more than two hidden layers are typically regarded as deep [157]. Including extra layers enables the perceptron to tackle nonlinear classification tasks effectively.

In the quest for the optimal architecture, there is no established theoretical foundation guiding the discovery or specification of such a design. The prevailing approach is one of trial and error, making it time-consuming and largely dependent on the expertise of the practitioner. Various methodologies can be employed, like 'rules of thumb', which determine the hidden layer's size by solely considering the number of elements in the input layer and the number of neurons in the output layer [158]. To highlight a few of these methodologies, we cite the pyramidal rule, which says the number of neurons in the hidden layer is determined by the product of parameter 'a' and the square root of the product of neurons in the input and output layers [159], with 'a' typically ranging between 0.5 and 2. In the average rule, the number of neurons in the hidden layer is equivalent to half the sum of neurons in the input and output layers [160], while the two-thirds rule stipulates that the number of neurons in the hidden layer is two-thirds of the neurons in the input layer plus the neurons in the output layer [161]. While these rules are straightforward to implement and offer quick outcomes, they are recognized as lacking a formal scientific basis [162]. A single-layer multilayer perceptron (MLP) with one hidden layer is usually sufficient for most applications [163,164]. The study examined four configurations, including one and two hidden layers with ten and fifty neurons. The optimal model was identified as having one hidden layer with fifty neurons.

The activation function processes the input vector to generate the total signal that a neuron receives [165]; in simple words, the activation function is used to activate and deactivate neurons. Additionally, Apicella et al. [166] show the activation function involves a two-step procedure for computing a neuron's output: initially, the neuron's input is determined through a functional mapping from R^d to R (typically a weighted sum), followed by computation of the neuron's output or activation using the activation function, which represents a functional mapping from R to R . Activation functions (AFs) play a significant role in neural networks by facilitating learning abstract features through non-linear transformations [167]. There are various activation functions used in neural networks, with some of the most popular ones including: sigmoid, hyperbolic tangent (tanh), rectified linear unit (ReLU), leaky ReLU, softmax, linear (identity), and Gaussian Error Linear Unit (GELU). Here, the following activation functions were tested: identity, softmax, ReLU, sigmoid, tanh. The linear function gives an output value equal to its input value. Using a linear activation function may not provide significant benefits, as it leads to a consistent gradient value throughout training iterations, and it can impede the network's ability to learn complex patterns from the data and improve error minimization. Therefore, linear functions are more suited for simpler tasks where complex pattern recognition is not a primary objective [168,169]. To introduce non-linearity into neural networks, logistic sigmoid and tanh activation functions are commonly used [170–173]. However, both the logistic sigmoid function and tanh tend to saturate for high and low inputs, resulting in the vanishing gradient issue [174]. This saturation means that the learning process can halt. As a result, the popularity of sigmoid and tanh activation functions has decreased, but they remain classic choices in neural network literature [175]. In our scenario, the best choice for the activation function was tanh.

With the multilayer perceptron (MLP), various solver parameters were investigated using three options: the limited-memory Broyden–Fletcher–Goldfarb–Shanno (lbfgs), stochastic gradient descent (sgd), and Adam solvers. The limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (lbfgs) employs approximated second-order gradient information, facilitating faster convergence towards the minimum and proving particularly effective for large-scale optimization problems, especially within a ‘one-vs-rest’ multiclass classification strategy [176–178]. Stochastic gradient descent (sgd) has emerged as a prominent method for optimizing contemporary models, iteratively reducing an objective function by calculating gradients from randomly sampled data [179]. The Adam algorithm [180] is widely adopted in machine learning as an optimization method, with its nomenclature derived from its focus on adaptive moments, amalgamating the advantages of the AdaGrad and RMSProp optimization algorithms. Adam is recognised as a first-order optimization method, noted for its ease of implementation, computational efficiency, and minimal memory requirements. Furthermore, it excels with non-stationary objectives and is well-suited for challenges presented by noisy or sparse gradients [181]. In our work, the lbfgs solver proved to be the most effective.

In multilayer perceptron (MLP) training, various learning rate schedules can be implemented, including constant, invscaling, and adaptive. The constant strategy’s learning rate remains unchanged throughout the training process. In the invscaling approach, the learning rate decreases gradually as training advances; it is inversely proportional to the square root of the number of iterations. Adaptive methods dynamically adjust the learning rate based on the gradient information acquired during training [182].

In our nitrate source scenario, the optimal learning rate was found to be constant, and the best model was achieved with a maximum of 5000 iterations and parameter alpha equal 10. The accuracy score reached 89.1 %, while the precision score stood at 89.0 %. The decision N source boundaries were well distinguished; however, they were not rectangular as presented in a Kendall box biplot.

Figure 5 shows the results of the decision tree algorithm for classifying nitrate N isotopic compositions among four sources. To obtain the source classification, several parameters needed to be set, including the splitter, which determines which feature and threshold are employed. When using the best option, the model selects the feature of highest importance; when using random selection, the model chooses features randomly but maintains the same distribution (for example, in Gini analysis, proline has an importance of 38 %, so it will be selected in 38 % of cases) [183]. In this study, the best option was the most accurate for N source classification tasks. The next parameter to be examined is ‘random state’. The ‘random state’ parameter determines which feature to select for a split if (and only if) there are two splits that are equally good (i.e. two features yield the exact same improvement in the selected splitting criteria). The default for this parameter is ‘None’, while any positive integer ensures that the tree will be generated using the same random seed each time it is trained. In this study, the random state was set to 1. Another parameter that can affect classification is ‘max depth’, which sets the maximum depth of the decision tree or tree in a random forest, indicating how deep the decision tree is from its root node [184, 185]. The preferred range for max depth is between 1 and 10 for smaller and more precise datasets. Theoretically, the maximum depth of a tree is equal to the number of training samples minus 1. Still, no algorithm will allow us to reach this point for obvious reasons, one being overfitting. In

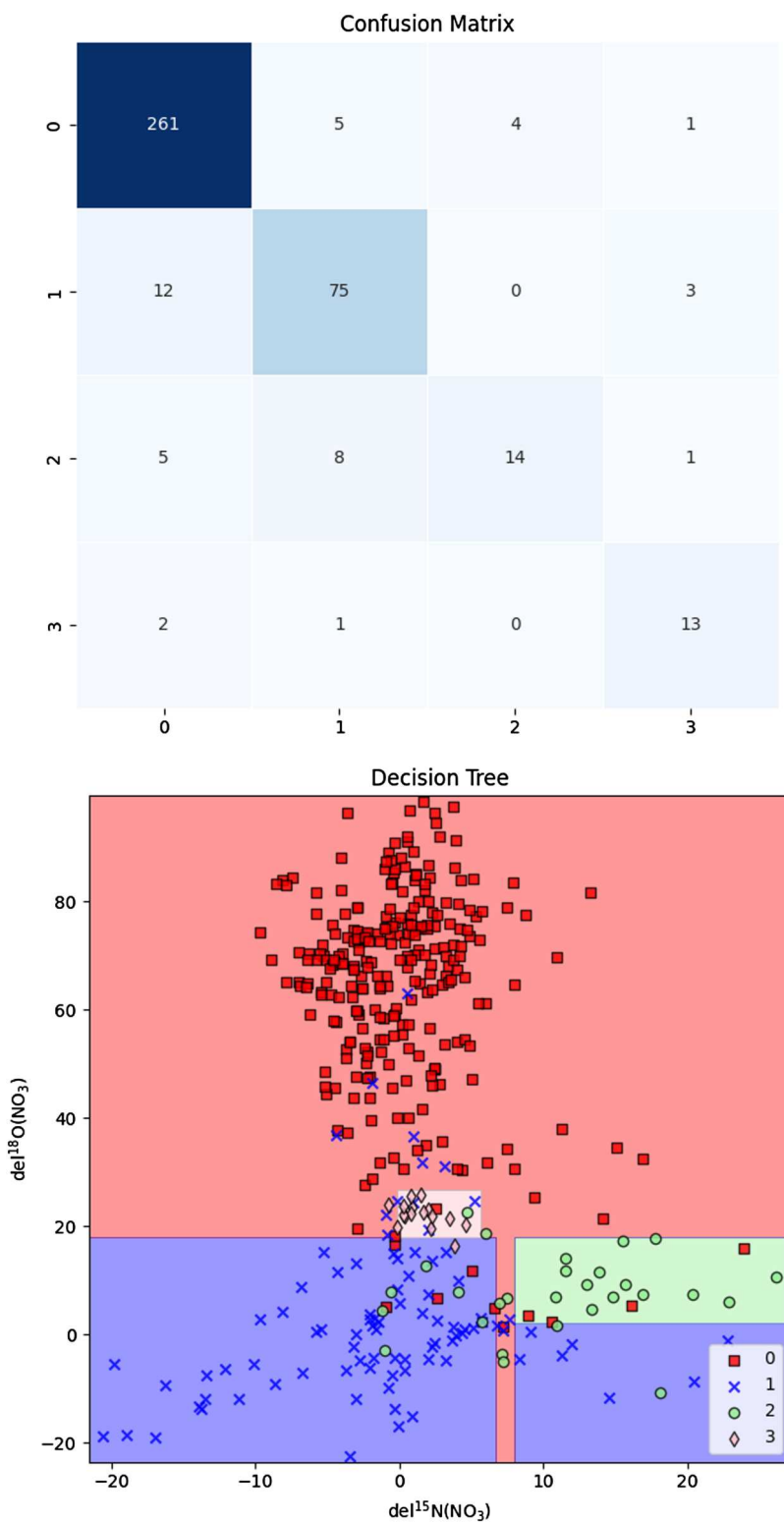


Figure 5. The results of the decision tree model for nitrate source classification use the same description for colours and markers as in the previous figures.

this study, the best option for max depth was determined to be 5. The hyperparameter 'min samples split' specifies the minimum number of samples essential for splitting an interior node [186]. In this case, the optimal integer for min samples split was also found to be 2. Another crucial hyperparameter is 'min samples leaf', which indicates the minimum number of data points required to create a leaf node. In this study, the best value for min samples leaf was determined to be 1. While computing the decision tree, the goal at each node is to select split conditions that best divide the dataset into homogeneous subsets. Therefore, the concept of the 'split criterion' was introduced, which is based on the notion of impurity, the function to measure the quality of a split. Supported criteria are 'gini' for the Gini impurity and 'log loss' and 'entropy' both for the Shannon information gain. The Gini index is a measure of total variance across the classes [187]. An alternative to the Gini index is entropy, which is a measure of the level of disorder in a subnode when objects are split based on the conditional attribute [188]. In this study, the best criterion for splitting was determined to be gini. Another parameter that was set is 'ccp_alpha', which was equal to 0.005. Finally, the N source classification was performed using the described parameters, yielding an accuracy score of 88.64% and a precision score of 89.6 %.

By utilizing the isotopic composition for nitrates, all three AI models can effectively categorize and visualize nitrate samples into four N sources. Consequently, the classification of new source samples can rely on the mathematical foundations extracted from machine learning models, thereby moving away from a dependence on subjective evaluations based on biplots and boxes. All AI classifications were characterized by high accuracy. Moreover, new, unknown data can be used in the future to improve the models. The application of machine learning algorithms offers a new alternative for the classification of nitrate and other sources based on their stable N and O isotope compositions. However, it is important to note no classification process is 'ideal', as no models consider mixing, past use of areas, and various other alteration processes (assumes conservative mixing). Nevertheless, AI can classify samples and identify decisive N source components for nitrate sources with greater precision and confidence. It is also worth mentioning that the accuracy of the models depends on the availability of high-quality and accurate source data. Progress in N isotopic studies, sharing of new source data, and applying new algorithms would enhance the identification and subsequent control of nitrate pollution.

4. Conclusions

Nitrate poses a significant threat of long-term contamination to surface and groundwater reservoirs, primarily due to intensive agriculture, inadequate wastewater management, and industrial activities. For water management and conservation strategies it is crucial to be able to accurately and confidently identify and delineate sources of N contamination to offer remediation approaches. Widespread N contamination is expected to escalate in various regions worldwide, underscoring the critical need to control the spread of pollution and eliminate its sources. The fundamental method of identifying nitrate sources in the environment relies on the dual (or triple) isotope approach, based on the N and O isotopic values of nitrogen and oxygen in nitrates. Fortunately, both isotopes can clearly differentiate between several nitrate sources, although some nitrate sources

exhibit significant overlap. The classic Kendall diagram to identify nitrates based on isotopic boxplots has been employed for three decades in almost unmodified form, and was developed based on limited data, with few data pertaining to atmospheric deposition, and the boundaries were judgmental due to the data constraints. For decades, the Kendall plot has been utilized in many environmental studies, often overlooking its limitations and without sufficient caution regarding the fact that many samples reflect the effects of mixing and other processes occurring under site- and source-specific conditions.

Since the 1990s, the number of studies and the increasing size of isotopic data have risen significantly. Moreover, more complex, defensible, and detailed tools are now available. Machine learning algorithms were employed here to better delineate the boundaries between sources of nitrates. These models achieved impressive accuracy (86–87 %) and produced visually appealing isotopic source plots for a dataset of approximately 1350 samples. Furthermore, the trained models presented here serve as tools to identify new samples and to subsequently improve the AI algorithm's performance. Thus, through machine learning models, one can assess the boundaries between N sources with high precision based on extensive data.

Acknowledgements

The authors acknowledge Carol Kendall and her USGS team for pioneering stable isotope work on nitrate. Their research on tracking nitrate sources has inspired other environmentalists over the decades.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- [1] Rockström J, Steffen W, Noone K, et al. A safe operating space for humanity. *Nature*. 2009;461(7263):472–475. doi:10.1038/461472a
- [2] Vitousek PM, Mooney HA, Lubchenco J, et al. Human domination of Earth's ecosystems. *Science*. 1997;277(5325):494–499. doi:10.1126/science.277.5325.494
- [3] Galloway JN, Aber JD, Erisman JW, et al. The nitrogen cascade. *Bioscience*. 2003;53(4):341–356. doi:10.1641/0006-3568(2003)053[0341:TNC]2.0.CO;2
- [4] Sebiló M, Mayer B, Nicolardot B, et al. Long-term fate of nitrate fertilizer in agricultural soils. *Proc Natl Acad Sci USA*. 2013;110(45):18185–18189. doi:10.1073/pnas.1305372110
- [5] FAO. World fertilizer trends and outlook to 2022. Rome: Food and Agriculture Organization of United Nations; 2020.
- [6] Martínez-Dalmau J, Berbel J, Ordóñez-Fernández R. Nitrogen fertilization. a review of the risks associated with the inefficiency of its use and policy responses. *Sustainability*. 2021;13(10):5625. doi:10.3390/su13105625

- [7] Ren X, Yue FJ, Tang J, et al. Nitrate transformation and source tracking of rivers draining into the Bohai Sea using a multi-tracer approach combined with an optimized Bayesian stable isotope mixing model. *J Hazard Mat.* **2024**;463:132901. doi:[10.1016/j.jhazmat.2023.132901](https://doi.org/10.1016/j.jhazmat.2023.132901)
- [8] Kharbush JJ, Robinson RS, Carter SJ. Patterns in sources and forms of nitrogen in a large eutrophic lake during a cyanobacterial harmful algal bloom. *Limnol Oceanogr.* **2023**;68(4):803–815. doi:[10.1002/lno.12311](https://doi.org/10.1002/lno.12311)
- [9] Singh S, Anil AG, Kumar V, et al. Nitrates in the environment: a critical review of their distribution, sensing techniques, ecological effects and remediation. *Chemosphere.* **2022**;287:131996. doi:[10.1016/j.chemosphere.2021.131996](https://doi.org/10.1016/j.chemosphere.2021.131996)
- [10] Essien EE, Said Abasse K, Côté A, et al. Drinking-water nitrate and cancer risk: a systematic review and meta-analysis. *Arch Environ Occup Health.* **2022**;77(1):51–67. doi:[10.1080/19338244.2020.1842313](https://doi.org/10.1080/19338244.2020.1842313)
- [11] Picetti R, Deeney M, Pastorino S, et al. Nitrate and nitrite contamination in drinking water and cancer risk: a systematic review with meta-analysis. *Environ Res.* **2022**;210:112988. doi:[10.1016/j.envres.2022.112988](https://doi.org/10.1016/j.envres.2022.112988)
- [12] Low AJ, Piper FJ. Nitrogen, sulphur and carbon uptake from some nitrogenous fertilizers using ^{15}N , ^{35}S and ^{13}C as tracers. *J Agric Sci.* **1957**;49(1):56–59. doi:[10.1017/S0021859600034316](https://doi.org/10.1017/S0021859600034316)
- [13] Bremner JM. Isotope-ratio analysis of nitrogen in nitrogen-15 tracer investigations. In: Norman AG, editor. *Methods of soil analysis: Part 2 Chemical and microbiological properties*. Madison, WI: American Society of Agronomy; **1965**. p. 1256–1286. (Agronomy; 9).
- [14] Kendall C. Tracing nitrogen sources and cycling in catchments. In: Kendall C, McDonnell JJ, editors. *Isotope tracers in catchment hydrology*. Amsterdam: Elsevier; **1998**. p. 519–576.
- [15] Mayer B. Assessing sources and transformations of sulphate and nitrate in the hydrosphere using isotope techniques. In: Aggarwal PK, Gat JR, Froehlich KFO, editors. *Isotopes in the water cycle: past, present and future of a developing science*. Dordrecht: Springer Netherlands; **2005**. p. 67–89.
- [16] Gebus B, Hałas S. Isotopic studies of nitrates—a short review. *Ann Univ Mariae Curie Skłodowska Phys.* **2016**;70:83.
- [17] Kendall C, Elliott EM, Wankel SD. Tracing anthropogenic inputs of nitrogen to ecosystems. In: Michener R, Lajtha K, editors. *Stable isotopes in ecology and environmental science*. Malden, MA: Blackwell; **2007**. p. 375–449.
- [18] Xue D, Botte J, De Baets B, et al. Present limitations and future prospects of stable isotope methods for nitrate source identification in surface- and groundwater. *Water Res.* **2009**;43(5):1159–1170. doi:[10.1016/j.watres.2008.12.048](https://doi.org/10.1016/j.watres.2008.12.048)
- [19] Lee KS, Lee DS, Lim SS, et al. Nitrogen isotope ratios of dissolved organic nitrogen in wet precipitation in a metropolis surrounded by agricultural areas in southern Korea. *Agric Ecosyst Environ.* **2012**;159:161–169. doi:[10.1016/j.agee.2012.07.010](https://doi.org/10.1016/j.agee.2012.07.010)
- [20] Liu XY, Xiao HW, Xiao HY, et al. Stable isotope analyses of precipitation nitrogen sources in Guiyang, southwestern China. *Environ Pollut.* **2017**;230:486–494. doi:[10.1016/j.envpol.2017.06.010](https://doi.org/10.1016/j.envpol.2017.06.010)
- [21] Shi Y, Li C, Jin Z, et al. Combining dual isotopes and a Bayesian isotope mixing model to quantify the nitrate sources of precipitation in Ningbo, East China. *Sci Total Environ.* **2021**;778:146297. doi:[10.1016/j.scitotenv.2021.146297](https://doi.org/10.1016/j.scitotenv.2021.146297)
- [22] Zeng J, Han G, Zhang S, et al. Nitrate dynamics and source identification of rainwater in Beijing during rainy season: insight from dual isotopes and Bayesian model. *Sci Total Environ.* **2023**;856:159234. doi:[10.1016/j.scitotenv.2022.159234](https://doi.org/10.1016/j.scitotenv.2022.159234)
- [23] Bateman AS, Kelly SD. Fertilizer nitrogen isotope signatures. *Isot Environ Health Stud.* **2007**;43(3):237–247. doi:[10.1080/10256010701550732](https://doi.org/10.1080/10256010701550732)
- [24] Lim SS, Lee SM, Lee SH, et al. Nitrogen isotope compositions of synthetic fertilizer, raw live-stock manure slurry, and composted livestock manure. *J Korean Soc Soil Sci Fert.* **2010**;43(4):453–457.

- [25] Chalk PM, Magalhães AM, Inácio CT. Towards an understanding of the dynamics of compost in the soil–plant–atmosphere system using ^{15}N tracer. *Plant Soil*. 2013;362:373–388. doi:10.1007/s11104-012-1358-5
- [26] Michalski G, Kolanowski M, Riha KM. Oxygen and nitrogen isotopic composition of nitrate in commercial fertilizers, nitric acid, and reagent salts. *Isot Environ Health Stud*. 2015;51(3):382–391. doi:10.1080/10256016.2015.1054821
- [27] Chang CC, Kendall C, Silva SR, et al. Nitrate stable isotopes: tools for determining nitrate sources among different land uses in the Mississippi river basin. *Can J Fish Aquat Sci*. 2002;59(12):1874–1885. doi:10.1139/f02-153
- [28] Kendall C, Aravena R. Nitrate isotopes in groundwater systems. In: Cook PG, Herczeg AL, editors. *Environmental tracers in subsurface hydrology*. Boston, MA: Springer US; 2000. p. 261–297.
- [29] Mayer B, Boyer EW, Goodale C, et al. Sources of nitrate in rivers draining sixteen watersheds in the northeastern US: isotopic constraints. *Biogeochemistry*. 2002;57:171–197. doi:10.1023/A:1015744002496
- [30] Deutsch B, Mewes M, Liskow I, et al. Quantification of diffuse nitrate inputs into a small river system using stable isotopes of oxygen and nitrogen in nitrate. *Org Geochem*. 2006;37(10):1333–1342. doi:10.1016/j.orggeochem.2006.04.012
- [31] Fenech C, Rock L, Nolan K, et al. The potential for a suite of isotope and chemical markers to differentiate sources of nitrate contamination: a review. *Water Res*. 2012;46(7):2023–2041. doi:10.1016/j.watres.2012.01.044
- [32] Pastén-Zapata E, Ledesma-Ruiz R, Harter T, et al. Assessment of sources and fate of nitrate in shallow groundwater of an agricultural area by using a multi-tracer approach. *Sci Total Environ*. 2014;470:855–864. doi:10.1016/j.scitotenv.2013.10.043
- [33] Yue FJ, Li SL, Liu CQ, et al. Tracing nitrate sources with dual isotopes and long term monitoring of nitrogen species in the Yellow River, China. *Sci Rep*. 2017;7(1):8537. doi:10.1038/s41598-017-08756-7
- [34] Jung H, Koh DC, Kim YS, et al. Stable isotopes of water and nitrate for the identification of groundwater flowpaths: A review. *Water (Basel)*. 2020;12(1):138. doi:10.3390/w12010138
- [35] Torres-Martínez JA, Mora A, Knappett PS, et al. Tracking nitrate and sulfate sources in groundwater of an urbanized valley using a multi-tracer approach combined with a Bayesian isotope mixing model. *Water Res*. 2020;182:115962. doi:10.1016/j.watres.2020.115962
- [36] Carrey R, Ballesté E, Blanch AR, et al. Combining multi-isotopic and molecular source tracking methods to identify nitrate pollution sources in surface and groundwater. *Water Res*. 2021;188:116537. doi:10.1016/j.watres.2020.116537
- [37] Lorette G, Sebilo M, Buquet D, et al. Tracing sources and fate of nitrate in multilayered karstic hydrogeological catchments using natural stable isotopic composition ($\delta^{15}\text{N-NO}_3^-$ and $\delta^{18}\text{O-NO}_3^-$). Application to the Toulon karst system (Dordogne, France). *J Hydrol*. 2022;610:127972. doi:10.1016/j.jhydrol.2022.127972
- [38] Li X, Lin J, Zhang L, et al. Hydrochemical evolution and nitrate source identification of river water and groundwater in Huashan watershed, China. *Sustainability*. 2024;16(1):423. doi:10.3390/su16010423
- [39] Venkiteswaran JJ, Boeckx P, Goody DC. Towards a global interpretation of dual nitrate isotopes in surface waters. *J Hydrol X*. 2019;4:100037.
- [40] Widory D, Petelet-Giraud E, Négrel P, et al. Tracking the sources of nitrate in groundwater using coupled nitrogen and boron isotopes: a synthesis. *Environ Sci Technol*. 2005;39(2):539–548. doi:10.1021/es0493897
- [41] Lasagna M, De Luca DA. Evaluation of sources and fate of nitrates in the western Po plain groundwater (Italy) using nitrogen and boron isotopes. *Environ Sci Pollut Res*. 2019;26:2089–2104. doi:10.1007/s11356-017-0792-6
- [42] Kruk MK, Mayer B, Nightingale M, et al. Tracing nitrate sources with a combined isotope approach ($\delta^{15}\text{N}_{\text{NO}_3}$, $\delta^{18}\text{O}_{\text{NO}_3}$ and $\delta^{11}\text{B}$) in a large mixed-use watershed in southern Alberta, Canada. *Sci Total Environ*. 2020;703:135043. doi:10.1016/j.scitotenv.2019.135043

- [43] Michalski G, Scott Z, Kabiling M, et al. First measurements and modeling of ^{17}O in atmospheric nitrate. *Geophys Res Lett*. 2003;30:14–11.
- [44] Morin S, Savarino J, Frey MM, et al. Comprehensive isotopic composition of atmospheric nitrate in the Atlantic Ocean boundary layer from 65°S to 79°N. *J Geophys Res*. 2009;114(D5):D05303.
- [45] Savarino J, Kaiser J, Morin S, et al. Nitrogen and oxygen isotopic constraints on the origin of atmospheric nitrate in coastal Antarctica. *Atmos Chem Phys*. 2007;7(8):1925–1945. doi:10.5194/acp-7-1925-2007
- [46] Bourgeois I, Savarino J, Caillon N, et al. Tracing the fate of atmospheric nitrate in a subalpine watershed using $\Delta^{17}\text{O}$. *Environ Sci Technol*. 2018;52(10):5561–5570. doi:10.1021/acs.est.7b02395
- [47] Ji X, Shu L, Li J, et al. Tracing nitrate sources and transformations using $\Delta^{17}\text{O}$, $\delta^{15}\text{N}$, and $\delta^{18}\text{O}-\text{NO}_3^-$ in a coastal plain river network of eastern China. *J Hydrol*. 2022;610:127829. doi:10.1016/j.jhydrol.2022.127829
- [48] Michalski G, Meixner T, Fenn M, et al. Tracing atmospheric nitrate deposition in a complex semiarid ecosystem using $\Delta^{17}\text{O}$. *Environ Sci Technol*. 2004;38(7):2175–2181. doi:10.1021/es034980+
- [49] Phillips DL, Gregg JW. Source partitioning using stable isotopes: coping with too many sources. *Oecologia*. 2003;136:261–269. doi:10.1007/s00442-003-1218-3
- [50] Parnell AC, Inger R, Bearhop S, et al. Source partitioning using stable isotopes: coping with too much variation. *PLoS One*. 2010;5(3):e9672. doi:10.1371/journal.pone.0009672
- [51] Parnell AC, Phillips DL, Bearhop S, et al. Bayesian stable isotope mixing models. *Environmetric*. 2013;24(6):387–399. doi:10.1002/env.2221
- [52] Stock BC, Jackson AL, Ward EJ, et al. Analyzing mixing systems using a new generation of Bayesian tracer mixing models. *PeerJ*. 2018;6:e5096. doi:10.7717/peerj.5096
- [53] Samborska K, Halas S. ^{34}S and ^{18}O in dissolved sulfate as tracers of hydrogeochemical evolution of the Triassic carbonate aquifer exposed to intense groundwater exploitation (Olkusz–Zawiercie region, southern Poland). *Appl Geochem*. 2010;25(9):1397–1414. doi:10.1016/j.apgeochem.2010.06.010
- [54] Samborska K, Halas S, Bottrell SH. Sources and impact of sulphate on groundwaters of Triassic carbonate aquifers, Upper Silesia, Poland. *J Hydrol*. 2013;486:136–150. doi:10.1016/j.jhydrol.2013.01.017
- [55] Mabit L, Gibbs M, Mbaye M, et al. Novel application of Compound Specific Stable Isotope (CSSI) techniques to investigate on-site sediment origins across arable fields. *Geoderma*. 2018;316:19–26. doi:10.1016/j.geoderma.2017.12.008
- [56] Wu X, Zheng XJ, Yin XW, et al. Seasonal variation in the groundwater dependency of two dominant woody species in a desert region of Central Asia. *Plant Soil*. 2019;444:39–55. doi:10.1007/s11104-019-04251-2
- [57] Sheng T, Yang PH, Xie GW, et al. [Nitrate-nitrogen pollution sources of an underground river in karst agricultural area using ^{15}N and ^{18}O isotope technique]. *Huan Jing Ke Xue*. 2018;39(10):4547–4555. Chinese.
- [58] Xu L, Jiang YJ, Duan SH, et al. [Quantification of nitrate sources to groundwater in karst trough-valley areas based on dual stable isotopes of $\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$ and the IsoSource model]. *Huan Jing Ke Xue*. 2020;41(8):3637–3645. Chinese.
- [59] Yang L, Han J, Xue J, et al. Nitrate source apportionment in a subtropical watershed using Bayesian model. *Sci Total Environ*. 2013;463:340–347. doi:10.1016/j.scitotenv.2013.06.021
- [60] Xia Y, Weller DE, Williams MN, et al. Using Bayesian hierarchical models to better understand nitrate sources and sinks in agricultural watersheds. *Water Res*. 2016;105:527–539. doi:10.1016/j.watres.2016.09.033
- [61] Li C, Li SL, Yue FJ, et al. Identification of sources and transformations of nitrate in the Xijiang River using nitrate isotopes and Bayesian model. *Sci Total Environ*. 2019;646:801–810. doi:10.1016/j.scitotenv.2018.07.345

- [62] Mao H, Wang G, Liao F, et al. Spatial variability of source contributions to nitrate in regional groundwater based on the positive matrix factorization and Bayesian model. *J Hazard Mat.* **2023**;445:130569. doi:[10.1016/j.jhazmat.2022.130569](https://doi.org/10.1016/j.jhazmat.2022.130569)
- [63] Wang J, Lu N, Fu B. Inter-comparison of stable isotope mixing models for determining plant water source partitioning. *Sci Total Environ.* **2019**;666:685–693. doi:[10.1016/j.scitotenv.2019.02.262](https://doi.org/10.1016/j.scitotenv.2019.02.262)
- [64] Matiatos I, Wassenaar LI, Monteiro LR, et al. Global patterns of nitrate isotope composition in rivers and adjacent aquifers reveal reactive nitrogen cascading. *Commun Earth Environ.* **2021**;2(1):52. doi:[10.1038/s43247-021-00121-x](https://doi.org/10.1038/s43247-021-00121-x)
- [65] Turing AM. On computable numbers, with an application to the Entscheidungsproblem. *Proc Lond Math Soc.* **1937**;s2-42:230–265. doi:[10.1112/plms/s2-42.1.230](https://doi.org/10.1112/plms/s2-42.1.230)
- [66] Wiener N. *Cybernetics, or control and communication in the animal and the machine.* 2nd ed. Cambridge, MA: MIT Press; **1948**.
- [67] Meng XL. Statistical paradises and paradoxes in big data (i) law of large populations, big data paradox, and the 2016 US presidential election. *Ann Appl Stat.* **2018**;12(2):685–726.
- [68] Crisci C, Ghattas B, Perera G. A review of supervised machine learning algorithms and their applications to ecological data. *Ecol Modell.* **2012**;240:113–122. doi:[10.1016/j.ecolmodel.2012.03.001](https://doi.org/10.1016/j.ecolmodel.2012.03.001)
- [69] Zhong S, Zhang K, Bagheri M, et al. Machine learning: new ideas and tools in environmental science and engineering. *Environ Sci Technol.* **2021**;55(19):12741–12754.
- [70] Ahmed AN, Othman FB, Afan HA, et al. Machine learning methods for better water quality prediction. *J Hydrol.* **2019**;578:124084. doi:[10.1016/j.jhydrol.2019.124084](https://doi.org/10.1016/j.jhydrol.2019.124084)
- [71] Yeom S, Giacomelli I, Fredrikson M, et al. Privacy risk in machine learning: analyzing the connection to overfitting. In: *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*. New York: Institute of Electrical and Electronics Engineers; **2018**. p. 268–282.
- [72] Ying X. An overview of overfitting and its solutions. *J Phys Conf Ser.* **2019**;1168:022022. doi:[10.1088/1742-6596/1168/2/022022](https://doi.org/10.1088/1742-6596/1168/2/022022)
- [73] Henderson P, Islam R, Bachman P, et al. Deep reinforcement learning that matters. In: *Proceedings of the 32nd AAAI Conference on Artificial Intelligence (AAAI-18)*; **2018**. p. 3207–3214. <https://ojs.aaai.org/index.php/AAAI/article/view/11694>
- [74] Rudin C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat Mach Intell.* **2019**;1(5):206–215. doi:[10.1038/s42256-019-0048-x](https://doi.org/10.1038/s42256-019-0048-x)
- [75] Linardatos P, Papastefanopoulos V, Kotsiantis S. Explainable AI: A review of machine learning interpretability methods. *Entropy* . **2020**;23(1):18. doi:[10.3390/e23010018](https://doi.org/10.3390/e23010018)
- [76] Sajedi-Hosseini F, Malekian A, Choubin B, et al. A novel machine learning-based approach for the risk assessment of nitrate groundwater contamination. *Sci Total Environ.* **2018**;644:954–962. doi:[10.1016/j.scitotenv.2018.07.054](https://doi.org/10.1016/j.scitotenv.2018.07.054)
- [77] Awais M, Aslam B, Maqsoom A, et al. Assessing nitrate contamination risks in groundwater: a machine learning approach. *Appl Sci.* **2021**;11(21):10034. doi:[10.3390/app112110034](https://doi.org/10.3390/app112110034)
- [78] Elzain HE, Chung SY, Senapathi V, et al. Comparative study of machine learning models for evaluating groundwater vulnerability to nitrate contamination. *Ecotox Environ Saf.* **2022**;229:113061. doi:[10.1016/j.ecoenv.2021.113061](https://doi.org/10.1016/j.ecoenv.2021.113061)
- [79] Matiatos I, Moeck C, Vystavna Y, et al. Nitrate isotopes in catchment hydrology: insights, ideas and implications for models. *J Hydrol.* **2023**;626:130326. doi:[10.1016/j.jhydrol.2023.130326](https://doi.org/10.1016/j.jhydrol.2023.130326)
- [80] Nelson DB, Basler D, Kahmen A. Precipitation isotope time series predictions from machine learning applied in Europe. *Proc Natl Acad Sci USA.* **2021**;118(26):e2024107118. doi:[10.1073/pnas.2024107118](https://doi.org/10.1073/pnas.2024107118)
- [81] Heydarizad M, Gimeno L, Minaei M, et al. Stable isotope signatures in Tehran's precipitation: insights from artificial neural networks, stepwise regression, wavelet coherence, and ensemble machine learning approaches. *Water (Basel).* **2023**;15(13):2357. doi:[10.3390/w15132357](https://doi.org/10.3390/w15132357)
- [82] Cemek B, Arslan H, Küçüktopcu E, et al. Comparative analysis of machine learning techniques for estimating groundwater deuterium and oxygen-18 isotopes. *Stoch Environ Res Risk Assess.* **2022**;36(12):4271–4285. doi:[10.1007/s00477-022-02262-7](https://doi.org/10.1007/s00477-022-02262-7)

- [83] Klein ES, Baltensperger AP, Welker JM. Complexity of Arctic Ocean water isotope ($\delta^{18}\text{O}$, $\delta^2\text{H}$) spatial and temporal patterns revealed with machine learning. *Elementa Sci Anthrop.* **2024**;12(1):00127. doi:[10.1525/elementa.2022.00127](https://doi.org/10.1525/elementa.2022.00127)
- [84] Yang Y, Shang X, Chen Z, et al. A support vector regression model to predict nitrate-nitrogen isotopic composition using hydro-chemical variables. *J Environ Manage.* **2021**;290:112674. doi:[10.1016/j.jenvman.2021.112674](https://doi.org/10.1016/j.jenvman.2021.112674)
- [85] Xie F, Cai G, Li G, et al. Basin-wide tracking of nitrate cycling in Yangtze River through dual isotope and machine learning. *Sci Total Environ.* **2024**;912:169656. doi:[10.1016/j.scitotenv.2023.169656](https://doi.org/10.1016/j.scitotenv.2023.169656)
- [86] Sebestyen SD, Kendall C, Elliott EM, et al. Nitrate isotope database for meteoric waters, surface waters, soil waters, and groundwaters. Fort Collins, CO: Forest Service Research Data Archive; **2019**.
- [87] Galdi P, Tagliaferri R, et al. Data mining: accuracy and error measures for classification and prediction. In: Ranganathan H, Gribskov M, Nakai K, editors. *Encyclopedia of bioinformatics and computational biology*, Vol. 1. Amsterdam: Elsevier; **2018**. p. 431–436.
- [88] Dinga R, Penninx BW, Veltman DJ, et al. Beyond accuracy: measures for assessing machine learning models, pitfalls and guidelines. *bioRxiv.* **2019**;743138:1–20. doi:[10.1101/743138](https://doi.org/10.1101/743138)
- [89] Powers DMW. Evaluation: from precision, recall and F-factor to ROC, informedness, markedness and correlation. *J MachLearn Technol.* **2011**;2:37–63. doi:[10.48550/arXiv.2010.16061](https://doi.org/10.48550/arXiv.2010.16061)
- [90] Gudivada V, Apon A, Ding J. Data quality considerations for big data and machine learning: going beyond data cleaning and transformations. *Int J Adv Softw.* **2017**;10(1):1–20.
- [91] Li P, Rao X, Blase J, et al. CleanML: A study for evaluating the impact of data cleaning on ml classification tasks. In: *2021 IEEE 37th International Conference on Data Engineering (ICDE)*. New York: Institute of Electrical and Electronics Engineers; **2021**. p. 13–24.
- [92] Budach L, Feuerpfeil M, Ihde N, et al. The effects of data quality on machine learning performance. *arXiv.* **2022**. doi:[10.48550/arXiv.2207.14529](https://doi.org/10.48550/arXiv.2207.14529)
- [93] Maharana K, Mondal S, Nemade B. A review: data pre-processing and data augmentation techniques. *Glob Transit Proc.* **2022**;3(1):91–99. doi:[10.1016/j.gltp.2022.04.020](https://doi.org/10.1016/j.gltp.2022.04.020)
- [94] Doherty K, Adams RG, Davey N. Non-Euclidean norms and data normalisation. In: *Proceedings of ESANN'2004 - European Symposium on Artificial Neural Networks*, Bruges (Belgium), 28–30 April 2004. Vreeland: De Facto Publications; **2004**. p. 181–186.
- [95] Mohamed MM, Schuller BW. Normalise for fairness: A simple normalisation technique for fairness in regression machine learning problems. *arXiv.* **2022**. doi:[10.48550/arXiv.2202.00993](https://doi.org/10.48550/arXiv.2202.00993)
- [96] Efron B, Tibshirani R. Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy. *Stat Sci.* **1986**;1(1):54–75.
- [97] Lahiri SN. Bootstrap methods: A review. In: Fan J, Koul HL, editors. *Frontiers in statistics*. Singapore: World Scientific; **2006**. p. 231–255.
- [98] Reed S, Lee H, Anguelov D, et al. Training deep neural networks on noisy labels with bootstrapping. *arXiv.* **2014**. doi:[10.48550/arXiv.1412.6596](https://doi.org/10.48550/arXiv.1412.6596)
- [99] Yap BW, Rani KA, Rahman HAA, et al. An application of oversampling, undersampling, bagging and boosting in handling imbalanced datasets. In: Tutut Herawan, Mustafa Mat Deris, Jemal Abawajy, editors. *Proceedings of the First International Conference on Advanced Data and Information Engineering (DaEng-2013)*. Singapore: Springer; **2014**. p. 13–22.
- [100] Shelke MS, Deshmukh PR, Shandilya VK. A review on imbalanced data handling using under-sampling and oversampling technique. *Int J Recent Trends Eng Res.* **2017**;3(4):444–449. doi:[10.23883/IJRTER.2017.3168.0UWXM](https://doi.org/10.23883/IJRTER.2017.3168.0UWXM)
- [101] Vrigazova B. The proportion for splitting data into training and test set for the bootstrap in classification problems. *Bus Syst Res.* **2021**;12(1):228–242. doi:[10.2478/bsrj-2021-0015](https://doi.org/10.2478/bsrj-2021-0015)
- [102] Efron B. Better bootstrap confidence intervals. *J Am Stat Assoc.* **1987**;82(397):171–185. doi:[10.1080/01621459.1987.10478410](https://doi.org/10.1080/01621459.1987.10478410)
- [103] Chernick MR. *Bootstrap methods: A guide for practitioners and researchers*. Hoboken, NJ: John Wiley & Sons; **2011**.

- [104] Michelucci U, Venturini F. Estimating neural network's performance with bootstrap: a tutorial. *Mach Learn Knowl Extr.* **2021**;3(2):357–373. doi:[10.3390/make3020018](https://doi.org/10.3390/make3020018)
- [105] Xiao HW, Xiao HY, Luo L, et al. Stable carbon and nitrogen isotope compositions of bulk aerosol samples over the South China Sea. *Atmos Environ.* **2018**;193:1–10. doi:[10.1016/j.atmosenv.2018.09.006](https://doi.org/10.1016/j.atmosenv.2018.09.006)
- [106] Fang YT, Koba K, Wang XM, et al. Anthropogenic imprints on nitrogen and oxygen isotopic composition of precipitation nitrate in a nitrogen-polluted city in southern China. *Atmos Chem Phys.* **2011**;11(3):1313–1325. doi:[10.5194/acp-11-1313-2011](https://doi.org/10.5194/acp-11-1313-2011)
- [107] Yang J, Hsu Y, Dai SC, et al. Isotopic composition of water-soluble nitrate in bulk atmospheric deposition at Dongsha Island: sources and implications of external N supply to the northern South China Sea. *Biogeosciences.* **2014**;11(7):1833–1846. doi:[10.5194/bg-11-1833-2014](https://doi.org/10.5194/bg-11-1833-2014)
- [108] Altieri KE, Hastings MG, Gobel AR, et al. Isotopic composition of rainwater nitrate at Bermuda: The influence of air mass source and chemistry in the marine boundary layer. *J Geophys Res Atmos.* **2013**;118(19):11–304.
- [109] Chmura W. Badania składu izotopowego azotanów w wodach podziemnych i powierzchniowych—aspekty metodyczne i zastosowania [PhD dissertation]. Kraków: Akademia Górniczo-Hutnicza; **2008**. Polish.
- [110] Wu Y, Zhang J, Liu S, et al. Nitrogen deposition in precipitation to a monsoon-affected eutrophic embayment: fluxes, sources, and processes. *Atmos Environ.* **2018**;182:75–86. doi:[10.1016/j.atmosenv.2018.03.037](https://doi.org/10.1016/j.atmosenv.2018.03.037)
- [111] Balint SJ, Joyce E, Pennino S, et al. Identifying sources and impacts of precipitation-derived nitrogen in Narragansett Bay, RI. *Estuar Coast.* **2022**;45(5):1287–1304. doi:[10.1007/s12237-021-01029-7](https://doi.org/10.1007/s12237-021-01029-7)
- [112] Kim Y, Lee I, Farquhar J, et al. Multi isotope systematics of precipitation to trace the sources of air pollutants in Seoul, Korea. *Environ Pollut.* **2021**;286:117548. doi:[10.1016/j.envpol.2021.117548](https://doi.org/10.1016/j.envpol.2021.117548)
- [113] Kelley CJ. Nitrate-nitrogen and nitrate-oxygen isotope ratios for identification of nitrate sources and dominant nitrogen cycle processes in a tile-drained dryland agricultural field [Doctoral dissertation]. Pullman, WA: Washington State University; **2011**.
- [114] Hastings MG, Steig EJ, Sigman DM. Seasonal variations in N and O isotopes of nitrate in snow at Summit, Greenland: implications for the study of nitrate in snow and ice cores. *J Geophys Res Atmos.* **2004**;109(D20):D20306. doi:[10.1029/2004JD004991](https://doi.org/10.1029/2004JD004991)
- [115] Felix JD, Elliott EM, Avery GB, et al. Isotopic composition of nitrate in sequential Hurricane Irene precipitation samples: implications for changing NO_x sources. *Atmos Environ.* **2015**;106:191–195. doi:[10.1016/j.atmosenv.2015.01.075](https://doi.org/10.1016/j.atmosenv.2015.01.075)
- [116] Hales HC, Ross DS, Lini A. Isotopic signature of nitrate in two contrasting watersheds of Brush Brook, Vermont, USA. *Biogeochemistry.* **2007**;84:51–66. doi:[10.1007/s10533-007-9074-6](https://doi.org/10.1007/s10533-007-9074-6)
- [117] Sanchez DA, Szyrkiewicz A, Faiia AM. Determining sources of nitrate in the semi-arid Rio Grande using nitrogen and oxygen isotopes. *Appl Geochem.* **2017**;86:59–69. doi:[10.1016/j.apgeochem.2017.09.012](https://doi.org/10.1016/j.apgeochem.2017.09.012)
- [118] Hu J, Liu W, Xing M, et al. Using a nitrogen and oxygen isotopic approach to identify nitrate sources and cycling in the Wei River of northwestern China. *Hum Ecol Risk Assess.* **2019**;25(3):755–771. doi:[10.1080/10807039.2018.1451744](https://doi.org/10.1080/10807039.2018.1451744)
- [119] Yue FJ, Liu CQ, Li SL, et al. Analysis of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ to identify nitrate sources and transformations in Songhua river, northeast China. *J Hydrol.* **2014**;519:329–339. doi:[10.1016/j.jhydrol.2014.07.026](https://doi.org/10.1016/j.jhydrol.2014.07.026)
- [120] Yu Q, Wang F, Li X, et al. Tracking nitrate sources in the Chaohu Lake, China, using the nitrogen and oxygen isotopic approach. *Environ Sci Pollut Res.* **2018**;25:19518–19529. doi:[10.1007/s11356-018-2178-9](https://doi.org/10.1007/s11356-018-2178-9)
- [121] Minet E, Coxon CE, Goodhue R, et al. Evaluating the utility of ^{15}N and ^{18}O isotope abundance analyses to identify nitrate sources: A soil zone study. *Water Res.* **2012**;46(12):3723–3736. doi:[10.1016/j.watres.2012.03.004](https://doi.org/10.1016/j.watres.2012.03.004)

- [122] Hall SJ, Weintraub SR, Bowling DR. Scale-dependent linkages between nitrate isotopes and denitrification in surface soils: implications for isotope measurements and models. *Oecologia*. 2016;181:1221–1231. doi:[10.1007/s00442-016-3626-1](https://doi.org/10.1007/s00442-016-3626-1)
- [123] Huang H, Liu M, Wang J, et al. Sources identification of nitrogen using major ions and isotopic tracers in Shenyang, China. *Geofluids*; 2018;2018. doi:[10.1155/2018/8683904](https://doi.org/10.1155/2018/8683904)
- [124] Shin WJ, Kim I, Ryu JS, et al. Dual isotopes of nitrate in Korean fertilizers and their application for identifying nitrate sources. *Episodes J Int Geosci*. 2019;42(4):313–319.
- [125] Shin WJ, Ryu JS, Mayer B, et al. Nitrogen, sulfur, and oxygen isotope ratios of animal- and plant-based organic fertilizers used in South Korea. *J Environ Qual*. 2017;46(3):559–567. doi:[10.2134/jeq2017.01.0018](https://doi.org/10.2134/jeq2017.01.0018)
- [126] Payet N, Nicolini E, Rogers K, et al. Evidence of soil pollution by nitrates derived from pig effluent using ^{18}O and ^{15}N isotope analyses. *Agron Sustain Dev*. 2010;30:743–751. doi:[10.1051/agro/2009056](https://doi.org/10.1051/agro/2009056)
- [127] Vitòria L, Otero N, Soler A, et al. Fertilizer characterization: isotopic data (N, S, O, C, and Sr). *Environ Sci Technol*. 2004;38(12):3254–3262. doi:[10.1021/es0348187](https://doi.org/10.1021/es0348187)
- [128] Grimm BL, Stern LA, Lowe AJ. Forensic utility of a nitrogen and oxygen isotope ratio time series of ammonium nitrate and its isolated ions. *Talanta*. 2018;178:94–101. doi:[10.1016/j.talanta.2017.08.105](https://doi.org/10.1016/j.talanta.2017.08.105)
- [129] Heaton TH, Stuart ME, Sapiano M, et al. An isotope study of the sources of nitrate in Malta's groundwater. *J Hydrol*. 2012;414:244–254. doi:[10.1016/j.jhydrol.2011.10.037](https://doi.org/10.1016/j.jhydrol.2011.10.037)
- [130] Roadcap GS, Hackley KC, Hwang HH. Application of nitrogen and oxygen isotopes to identify sources of nitrate. In: *Proceedings, 12th Annual Illinois Groundwater Consortium Symposium*. Makanda, IL: Illinois Groundwater Consortium; 2002. <https://core.ac.uk/download/pdf/29157559.pdf>
- [131] Kazakis N, Matiatos I, Ntona MM, et al. Origin, implications and management strategies for nitrate pollution in surface and ground waters of Anthemountas basin based on a $\delta^{15}\text{N}\text{-NO}_3^-$ and $\delta^{18}\text{O}\text{-NO}_3^-$ isotope approach. *Sci Total Environ*. 2020;724:138211. doi:[10.1016/j.scitotenv.2020.138211](https://doi.org/10.1016/j.scitotenv.2020.138211)
- [132] Kotsiantis SB, Zaharakis I, Pintelas P. Supervised machine learning: A review of classification techniques. In: Maglogiannis IG, editor. *Emerging artificial intelligence applications in computer engineering*. Amsterdam: IOS Press; 2007. p. 3–24. (Frontiers in artificial intelligence and applications; 160).
- [133] Musa AB. Logistic regression classification for uncertain data. *Res J Math Stat Sci*. 2014;2320:6047.
- [134] Wong TT. Alternative prior assumptions for improving the performance of naïve Bayesian classifiers. *Data Min Knowl Discov*. 2009;18:183–213. doi:[10.1007/s10618-008-0101-6](https://doi.org/10.1007/s10618-008-0101-6)
- [135] Vedaraj M, Anita CS, Muralidhar A, et al. Early prediction of lung cancer using Gaussian Naive Bayes classification algorithm. *Int J Intell Syst Appl Eng*. 2023;11(6s):838–848.
- [136] Ren J, Lee SD, Chen X, et al. Naive Bayes classification of uncertain data. In: 2009 9th IEEE International Conference on Data Mining. New York: Institute of Electrical and Electronics Engineers; 2009. p. 944–949. <https://www.computer.org/csdl/proceedingsarticle/icdm/2009/3895z004/12OmNzgwmPP>
- [137] Qin B, Xia Y, Wang S, et al. A novel Bayesian classification for uncertain data. *Knowl Based Syst*. 2011;24(8):1151–1158. doi:[10.1016/j.knosys.2011.04.011](https://doi.org/10.1016/j.knosys.2011.04.011)
- [138] de Holanda Maia MR, Plastino A, Freitas AA. An ensemble of Naive Bayes classifiers for uncertain categorical data. In: 2021 IEEE International Conference on Data Mining (ICDM). New York: Institute of Electrical and Electronics Engineers; 2021. p. 1222–1227. <https://www.proceedings.com/content/062/062332webtoc.pdf>
- [139] Huang Y, Li L. Naive Bayes classification algorithm based on small sample set. In: Ling L, editor. 2011 IEEE International Conference on Cloud Computing and Intelligence Systems. New York: Institute of Electrical and Electronics Engineers; 2011. p. 34–39.
- [140] Sangounpao K, Muenchaisri P. Ontology-based Naive Bayes short text classification method for a small dataset. In: Masahide Nakamura, Hiroaki Hirata, Takayuki Ito, et al., editors. 2019 20th IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence,

- Networking and Parallel/Distributed Computing (SNPD). New York: Institute of Electrical and Electronics Engineers; 2019. p. 53–58.
- [141] Rosenblatt F. Two theorems of statistical separability in the perceptron. In: Mechanization of thought processes, symposium held at the national physical laboratory. London: HM Stationary Office; 1959. p. 421–456. <https://archive.org/details/mechanisationoft02nati/mode/2up>
- [142] da Silva AP, Chiari LP, Guimaraes AR, et al. Drug design of new 5-HT6R antagonists aided by artificial neural networks. *J Mol Graph Model*. 2021;104:107844. doi:10.1016/j.jmgm.2021.107844
- [143] Choi RY, Coyner AS, Kalpathy-Cramer J, et al. Introduction to machine learning, neural networks, and deep learning. *Transl Vis Sci Technol*. 2020;9(2):14.
- [144] Panchal G, Ganatra A, Kosta YP, et al. Behaviour analysis of multilayer perceptrons with multiple hidden neurons and hidden layers. *Int J Comput Theory Eng*. 2011;3(2):332–337. doi:10.7763/IJCTE.2011.V3.328
- [145] Garbin C, Zhu X, Marques O. Dropout vs. batch normalization: an empirical study of their impact to deep learning. *Multimed Tools Appl*. 2020;79(19):12777–12815. doi:10.1007/s11042-019-08453-9
- [146] Abdolrasol MG, Hussain SS, Ustun TS, et al. Artificial neural networks based optimization techniques: a review. *Electronics (Basel)*. 2021;10(21):2689. doi:10.3390/electronics10212689
- [147] Granziol D, Zohren S, Roberts S. Learning rates as a function of batch size: a random matrix theory approach to neural network training. *J Mach Learn Res*. 2022;23(173):1–65.
- [148] Rai K, Devi MS, Guleria A. Decision tree based algorithm for intrusion detection. *Int J Adv Netw Appl*. 2016;7(4):2828.
- [149] Shamrat FJM, Tasnim Z, Ghosh P, et al. Personalization of job circular announcement to applicants using decision tree classification algorithm. In: 2020 IEEE International Conference for Innovation in Technology (INOCON). New York: Institute of Electrical and Electronics Engineers; 2020. p. 1–5.
- [150] Purdilă V, Pentiuc ŞG. Fast decision tree algorithm. *Adv Electr Comput Eng*. 2014;14(1):65–68. doi:10.4316/AECE.2014.01010
- [151] Priyam A, Abhijeeta GR, Rathee A, et al. Comparative analysis of decision tree classification algorithms. *Int J Curr Eng Technol*. 2013;3(2):334–337.
- [152] Bhargava N, Sharma G, Bhargava R, et al. Decision tree analysis on J48 algorithm for data mining. *Int J Adv Res Comput Sci Softw Eng*. 2013;3(6):1114–1119.
- [153] Bansal M, Goyal A, Choudhary A. A comparative analysis of k-nearest neighbor, genetic, support vector machine, decision tree, and long short term memory algorithms in machine learning. *Decis Anal J*. 2022;3:100071. doi:10.1016/j.dajour.2022.100071
- [154] Graves A, Mohamed AR, Hinton G. Speech recognition with deep recurrent neural networks. In: Michael Adams, Vicky Zhao, editors. 2013 IEEE International Conference on Acoustics, Speech and Signal Processing. New York: Institute of Electrical and Electronics Engineers; 2013. p. 6645–6649.
- [155] Benardos PG, Vosniakos GC. Optimizing feedforward artificial neural network architecture. *Eng Appl Artif Intell*. 2007;20(3):365–382. doi:10.1016/j.engappai.2006.06.005
- [156] Hunter D, Yu H, Pukish MS, III, et al. Selection of proper neural network sizes and architectures —A comparative study. *IEEE Trans Ind Inform*. 2012;8(2):228–240. doi:10.1109/TII.2012.2187914
- [157] Bengio Y. Learning deep architectures for AI. *Found Trends Mach Learn*. 2009;2(1):1–127. doi:10.1561/22000000006
- [158] Ke J, Liu X. Empirical analysis of optimal hidden neurons in neural network modeling for stock prediction. In: Randall Bilof, editor. 2008 IEEE Pacific-Asia Workshop on Computational Intelligence and Industrial application, Vol. 2. New York: Institute of Electrical and Electronics Engineers; 2008. p. 828–832.
- [159] Palit AK, Popovic D. Computational intelligence in time series forecasting: theory and engineering applications. Heidelberg: Springer Science & Business Media; 2006.

- [160] Thomas AJ, Petridis M, Walters SD, et al. On predicting the optimal number of hidden nodes. In: Juan E. Guerrero, editor. 2015 International Conference on Computational Science and Computational Intelligence (CSCI). New York: Institute of Electrical and Electronics Engineers; 2015. p. 565–570.
- [161] Panchal FS, Panchal M. Review on methods of selecting number of hidden nodes in artificial neural network. *Int J Comput Sci Mobile Comput*. 2014;3(11):455–464.
- [162] Majalca R, Acosta PR. Convex hulls and the size of the hidden layer in a MLP based classifier. *IEEE Lat Am Trans*. 2019;17(06):991–999. doi:10.1109/TLA.2019.8896822
- [163] Guliyev NJ, Ismailov VE. A single hidden layer feedforward network with only one neuron in the hidden layer can approximate any univariate function. *Neural Comput*. 2016;28(7):1289–1304. doi:10.1162/NECO_a_00849
- [164] Chen S, Klivans AR, Meka R. Efficiently learning any one hidden layer ReLU network from queries. *Adv Neural Inf Process Syst*. 2021;34:24087–24098.
- [165] Duch W, Jankowski N. Transfer functions: hidden possibilities for better neural networks. In: ESANN'2001 Proceedings - European Symposium on Artificial Neural Networks, Bruges (Belgium), 25–27 April 2001. Vreeland: De Facto Publications; 2001. p. 81–94.
- [166] Apicella A, Donnarumma F, Isgrò F, et al. A survey on modern trainable activation functions. *Neural Netw*. 2021;138:14–32. doi:10.1016/j.neunet.2021.01.026
- [167] Duch W, Jankowski N. Survey of neural transfer functions. *Neural Comput Surv*. 1999;2(1):163–212.
- [168] Wanto A, Windarto AP, Hartama D, et al. Use of binary sigmoid function and linear identity in artificial neural networks for forecasting population density. *Int J Inf Syst Technol*. 2017;1(1):43–54.
- [169] Manessi F, Rozza A. Learning combinations of activation functions. In: Daniel Lopresti, Ran He, editors. 2018 24th International Conference on Pattern Recognition (ICPR). New York: Institute of Electrical and Electronics Engineers; 2018. p. 61–66.
- [170] Pratiwi H, Windarto AP, Susliansyah S, et al. Sigmoid activation function in selecting the best model of artificial neural networks. *J Phys Conf Ser*. 2020;1471(1):012010. doi:10.1088/1742-6596/1471/1/012010
- [171] Szandala T, et al. Review and comparison of commonly used activation functions for deep neural networks. In: Bhoi AK, Mallick PK, Liu C-M, editors. Bio-inspired neurocomputing. Singapore: Springer; 2021. p. 203–224.
- [172] Kumar A, Sodhi SS. Some modified activation functions of hyperbolic tangent (TanH) activation function for artificial neural networks. In: Abhishek Bhattacharya, Soumi Dutta, Paramartha Dutta, et al., editors. International Conference on Innovations in Data Analytics. Singapore: Springer Nature; 2022. p. 369–392.
- [173] Youn YR, Hong JK. Enhancing convolutional neural network performance through optimized sigmoid activation function modeling. *Asia-Pac J Conv Res Interch*. 2023;9(10):51–59.
- [174] Dubey SR, Singh SK, Chaudhuri BB. Activation functions in deep learning: A comprehensive survey and benchmark. *Neurocomputing*. 2022;503:92–108. doi:10.1016/j.neucom.2022.06.111
- [175] Mercioni MA, Holban S. The most used activation functions: classic versus current. In: 2020 International Conference on Development and Application Systems (DAS). New York: Institute of Electrical and Electronics Engineers; 2020. p. 141–145.
- [176] Schraudolph NN, Yu J, Günter S. A stochastic quasi-newton method for online convex optimization. In: Meila M, Shen X, editors. Proceedings of the Eleventh International Conference on Artificial Intelligence and Statistics. 2007. p. 436–443. (PMLR; 2).
- [177] Fan RE, Chang KW, Hsieh CJ, et al. LIBLINEAR: A library for large linear classification. *J Mach Learn Res*. 2008;9:1871–1874.
- [178] Gupta A, Parmar R, Suri P, et al. Determining accuracy rate of artificial intelligence models using Python and R-Studio. In: 2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N). New York: Institute of Electrical and Electronics Engineers; 2021. p. 889–894.

- [179] Yang X. Stochastic gradient variance reduction by solving a filtering problem. arXiv. 2020. doi:[10.48550/arXiv.2012.12418](https://doi.org/10.48550/arXiv.2012.12418)
- [180] Kingma DP, Ba J. Adam: A method for stochastic optimization. In: Yoshua Bengio, Yann LeCun, editors. Proceedings of the 3rd International Conference on Learning Representations (ICLR). San Diego, CA, USA; 2015.
- [181] Feng X, Ma G, Su SF, et al. A multi-layer perceptron approach for accelerated wave forecasting in Lake Michigan. *Ocean Eng.* 2020;211:107526. doi:[10.1016/j.oceaneng.2020.107526](https://doi.org/10.1016/j.oceaneng.2020.107526)
- [182] Brunese L, Mercaldo F, Reginelli A, et al. Neural networks for lung cancer detection through radiomic features. In: 2019 International Joint Conference on Neural Networks (IJCNN). New York: Institute of Electrical and Electronics Engineers; 2019. p. 1–10.
- [183] Truica CO, Leordeanu CA. Classification of an imbalanced data set using decision tree algorithms. *UPB Sci Bull C: Electr Eng Comput Sci.* 2017;79:69–84.
- [184] Patange AD, Pardeshi SS, Jegadeeshwaran R, et al. Augmentation of decision tree model through hyper-parameters tuning for monitoring of cutting tool faults based on vibration signatures. *J Vib Eng Technol.* 2023;11(8):3759–3777. doi:[10.1007/s42417-022-00781-9](https://doi.org/10.1007/s42417-022-00781-9)
- [185] Oktafiani R, Hermawan A, Avianto D. Max depth impact on heart disease classification: decision tree and random forest. *Jurnal RESTI.* 2024;8(1):160–168.
- [186] Van Rijn JN, Hutter F. Hyperparameter importance across datasets. In: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. 2018: 2367–2376.
- [187] James G, Witten D, Hastie T, et al. Tree-based methods. In: James G, Witten D, Hastie T, et al., editors. An introduction to statistical learning: with applications in Python. Cham: Springer International Publishing; 2023. p. 331–366.
- [188] Aning S, Przybyła-Kasperek M. Comparative study of twoing and entropy criterion for decision tree classification of dispersed data. *Procedia Comput Sci.* 2022;207:2434–2443. doi:[10.1016/j.procs.2022.09.301](https://doi.org/10.1016/j.procs.2022.09.301)
- [189] Kendall C, Young M, Silva SR, et al. Tracing nutrient and organic matter sources and biogeochemical processes in the sacramento river and northern delta: proof of concept using stable isotope data. Data Release. Reston, VI: US Geological Survey; 2015. (Data Release; 110).